

Question bank module 5

1. State and prove Nyquist criterion for zero ISI
2. With a neat diagram explain the timing features pertaining to eye diagram and its interpretation for base band binary data transmission.
3. Define M-ary . Obtain the constellation of QAM for $M=4$ and draw the signal space diagram
4. Define ISI. With a neat block diagram Explain the digital PAM transmission through bandlimited baseband channels and obtain the expression for ISI.
5. What is eye pattern? What is the Nyquist criterion for zero ISI? Given an example of the pulse with zero ISI
6. Describe the Nyquist criterion for distortionless baseband transmission find out the ideal solution for zero ISI.
7. Define Signal-to-Noise Ratio (SNR). Explain the different types of external and internal noise.
8. Develop a code to generate and plot eye diagram
9. Illustrate the concept of noise in cascaded stages with a diagram. Write Friis' formula and mention its terms.
10. Develop a code to generate NRZ and RZ pulse.
11. Develop a code to generate and plot eye diagram.